Sri Lanka Institute of Advanced Technological Education

HNDIT2062 – ICT PROJECT (GROUP)

PROJECT PROPOSAL

HOSTEL MANAGEMENT SYSTEM

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Hostel Management System

1. Problem Statement

The hostel at the Advanced Technological Institute, Kurunegala, accommodates a large number of students who require proper management of block allocations, student records, and hostel payments. Currently, many hostel administrative tasks are handled manually using paper-based records or simple spreadsheets.

These traditional methods often lead to problems such as data duplication, inaccurate records, and difficulty in tracking block availability. Managing student accommodation and monthly hostel payments manually becomes time-consuming and inefficient, especially as the number of students increases.

Additionally, administrators face challenges in maintaining up-to-date records of students staying in the hostel and responding quickly during emergencies due to the lack of a centralized digital system.

Therefore, an automated Hostel Management System is required for the hostel at the Advanced Technological Institute, Kurunegala, to efficiently manage student accommodation, block allocations, booking records, and payment details while providing quick access to accurate information.

2. Objectives and Goals

The main objectives of this project are:

- To design and develop a computerized hostel management system to automate and simplify hostel operations.

Specific Objectives:

- To manage student registration details efficiently.
- To manage block reservation and availability.
- To track the number of students in each block.
- To monitor student attendance (in/out status).
- To manage facility fee management.
- To provide secure access for Director and Warden.
- To send emergency notifications to students via email.
- To improve efficiency and reduce manual errors.

3. Proposed Solution / Approach

The proposed solution is to develop a Hostel Management System using Java programming language and MySQL database.

The system will provide a user-friendly interface that allows authorized users (Director and Warden) to manage hostel operations efficiently. The system will include modules such as login management, student management, block management, payment management, and notification management.

The system will allow users to:

- Add, update, delete, and search student records
- Allocate blocks and monitor block occupancy
- Check available blocks and remaining capacity
- Track whether a student is currently present in the hostel
- Record and manage monthly payments of students
- Facility fee
- Generate simple payment reports
- Send email notifications during emergency situations

The graphical user interface will be developed using Java Swing or JavaFX, and JDBC will be used to connect the system with the MySQL database.

4. Scope and Limitations

Scope

The system will cover:

- Management of student information.
- Management of hostel blocks (Boys and Girls hostels).
- Block allocation (50 blocks per hostel, 4 students per block).
- Tracking block occupancy and availability.
- Monitoring student presence (in/out status).
- Facility fee.
- Sending emergency email notifications.
- Role-based access (Director and Warden).

Limitations

The limitations of this system include:

- The system will only be accessible to authorized staff (Director and Warden).
- Students cannot access the system directly.
- Online payment gateway will not be implemented (manual payment recording only).
- Mobile application support will not be implemented.
- The system is designed for a single campus environment.

5. Methodology

The project will follow a structured Software Development Life Cycle (SDLC):

i. **Requirement Analysis**

Identify and analyze system requirements including student management, block allocation, payment handling, and notification features.

ii. **System Design**

Design system architecture and create diagrams such as Use Case Diagrams and ER Diagrams.

iii. **Database Design**

Design MySQL database with tables such as Students, Blocks, Allocations, Payments, and Attendance.

iv. **System Implementation**

Develop the system using Java, with GUI using Java Swing/JavaFX and database connectivity using JDBC.

v. **Testing**

Test all system functionalities and fix errors to ensure reliability.

vi. **Documentation and Deployment**

Finally, system documentation will be prepared, and the system will be demonstrated as part of the project presentation.

6. Timeline and Milestones (15 Weeks)

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14	Week 15
Project Planning															
Requirement Analysis															
System Design / Wireframe															
Database Design															
Development Phase 1															
Development Phase 2															
Integration															
Testing															
Improvement															
Documentation															
Presentation															

7. Resources and Budget

Resources

- Laptop
- Internet connection for research
- Java Development Environment (NetBeans)
- MySQL Database Server

Software Tools

- Java Development Kit (JDK)
- MySQL Database
- MySQL Workbench
- JavaFX for GUI development

Budget

The project uses open-source technologies; therefore, the cost is minimal.

8. Literature Review

Hostel management systems are widely used to automate accommodation and payment-related operations. Manual systems are inefficient and prone to errors, especially when managing large numbers of students and tracking payments.

Modern systems use database technologies to store and manage data efficiently. Java and MySQL are commonly used due to their reliability and scalability. Studies show that automated management systems improve efficiency, reduce human errors, and provide better control over hostel operations and payments.